

Turing Degrees

A Unified Research Report

Open problems, proposed tower reductions,
and the cost of paper-and-Lean verification

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Abstract

Three research reports are integrated into one catalogue of thirty-nine topics and three separately identified bounded research programmes. Shared definitions, feasibility assessments, source records, and formalization requirements are consolidated; distinct variants are retained rather than silently identified. The most consequential difference between the inputs concerns towers: one report proposes elementary strong reductions under explicit computable-column conventions, while another treats the same comparisons as transformation-search targets. We preserve the complete proposed proofs and their diagnostic lemmas, but separate mathematical claims under those conventions from external acceptance, novelty, and the interpretation of the published questions. The report also combines metric geometry, coarse and effective-dense representatives, local oracle classes, lattice spectra, categoricity, randomness, dimension, and global structure. A unified Lean audit distinguishes inherited source inspection from successful compilation, and a crosswalk accounts for every original numbered entry.

This is a synthesis of the supplied reports, not a claim of thirty-nine independent open conjectures or a new, exhaustive literature review. No Lean development is claimed to have been compiled. The included finite checks do not certify the infinite mathematical assertions.

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1 Executive synthesis and editorial scope

1.1 The three inputs and the integrated result

The supplied archive contains three reports, each dated 17 September 2026. We call them **Report A**, the research map and feasibility audit; **Report B**, the research portfolio with proposed tower constructions; and **Report C**, the broader research agenda with diagnostic lemmas. Their full titles and original paths are recorded in the bibliography and the crosswalk in Appendix A.[1, 2, 3]

Each input has twenty numbered research entries, but the lists overlap substantially. The unified catalogue has **39 topics**; it splits the distinct classical and relative dimension questions that one original entry grouped together. It also preserves three bounded programmes from Report A without counting them as established literature-open problems. Some topics contain refinements, restricted versions, or linked conjectures. Thus neither the original sixty entries nor the unified thirty-nine topics are counts of logically independent conjectures.

The organizing principle is the intended mathematical statement, not the original section order. Common foundations and software cautions appear once. Detailed constructions and proofs are retained. Where assessments differed, the distinction is explained instead of averaging incompatible labels. All original entries remain traceable through the crosswalk and the accompanying CSV.

1.2 Two research tracks, not one merged recommendation

Validation track: T1–T4. Report B supplies two explicit padding arguments proposing the stronger conclusion

$$\mathcal{T} \equiv_s \mathcal{T}^- \equiv_s \mathcal{TF}.$$

Report C identifies defects in two simpler approaches, but those defects are addressed by the finite-fiber and moving-cutoff features of the padding constructions. The appropriate next step is therefore to audit the proposed proofs, their exact input conventions, their relationship to the source questions, and their novelty—not to ignore them and restart an unconstrained separation search. The proofs are retained in Section 4. They have not been independently reviewed, accepted by the source author, or checked in Lean.

Unresolved-target track: M1, M3, and C1. For a genuinely unresolved bounded attempt, the combined reports favor the five-cycle metric realization, the diameter-three-versus-four question, and a counterexample to least Turing degrees in nonuniform coarse classes. These are speculative choices, not forecasts. A short conventional argument might exist; a complete formal proof could still require a substantial foundation. The local questions Q1–Q3, the effective-dense questions C2–C3, and a sharply defined categoricity subcase are exploratory alternatives. Programmes R1–R3 are more controllable only after fixing a genuinely new, finite or restricted target.

1.3 Status evidence and feasibility are different axes

The catalogue separates the following evidence descriptions. **O26**, **O25**, **O24**, and **O21** mean an explicit open-question statement or listing of that date inherited from the reports. **S07**, **S12**, and **S17** flag older primary-source leads for which current specialist reconfirmation is particularly important. **Ref** identifies a refinement or a precise component of a broader source question. **V** flags a supplied proof proposal rather than externally established closure of an open problem. A recent preprint is not silently treated as a peer-reviewed publication.

The unified feasibility tiers are:

V: validation candidate.

A detailed proposed proof is available. Its semantics, correctness, novelty, and formal realization still require separate validation.

B: bounded speculative target.

A specific unresolved statement could conceivably admit a short argument. No success probability is asserted.

E: exploratory target.

A concrete direction exists, but construction or formalization risk is appreciably larger. A sharply stated restriction may be a better first project.

H: high-dependency full target.

The known approaches or missing foundations make this a poor default short end-to-end project, without implying any mathematical impossibility.

R: bounded research programme.

The restriction or auxiliary theorem itself must first be specified and checked for novelty; success need not settle the parent problem.

The original four-to-eight-week horizons are planning assumptions for a specialist with an effective Lean workflow, not completion commitments. Mathematical elegance, freshness of a problem listing, and formal tractability are three different considerations.

1.4 What was checked for this unified edition

The complete LaTeX sources of all three reports were used. This edition normalizes notation and identifiers, deduplicates the bibliography, and resolves internal discrepancies without asserting that all inherited literature and software audits have been repeated. Targeted source checks on 18 September 2026 confirmed the displayed tower questions and the weak-tower wording in McDonald’s July preprint, Gerdes’s least-representative and set-representative questions, Zhao’s 16 September dispersive/AEWG status discussion, and the publication details of Marks–Slaman–Steel.[4, 5, 6, 7]

In particular, McDonald’s weak-tower definition does not repeat the infinitude and computability requirements in its wording; this report makes those requirements explicit rather than pretending that the interpretive issue is settled. Zhao’s account reports a recent attractive-degree characterization, whereas the dispersive-to-AEWG implication remains a question. The bibliography corrects the Marks–Slaman–Steel page range to 200–219. The original software evidence remains a *dated source audit*, not a current clean-build certificate.[4, 6, 7, 1, 2, 3]

2 Common mathematical language and representation boundaries

For $A, B \subseteq \mathbb{N}$, write $A \leq_T B$ when an oracle program using the total characteristic function of B computes that of A . Mutual reducibility is $A \equiv_T B$, and

$$\mathcal{D} = \mathcal{P}(\mathbb{N})/\equiv_T, \quad \deg(A) \leq \deg(B) \iff A \leq_T B.$$

The least degree $\mathbf{0}$ consists of computable sets; the join is induced by $A \oplus B = \{2n : n \in A\} \cup \{2n+1 : n \in B\}$. The jump A' is the diagonal halting problem relative to A . The c.e. degrees $\mathcal{R}$ and the d.c.e. degrees $\mathcal{D}_2$ are different suborders of $\mathcal{D}$; a d.c.e. set is a difference of two c.e. sets. A meet must be taken in its stated ambient order, and a property of one representative need not hold of every equivalent representative.

For sets, $A \subseteq^* B$ means $A \setminus B$ is finite. For total functions, $f \leq^* g$ means $\{x : f(x) > g(x)\}$ is finite, and $f <^\infty g$ means $\{x : f(x) < g(x)\}$ is infinite. *Unbounded* means $\forall b \exists x f(x) > b$; it does not mean that $f(x)$ tends to infinity. Total numerical functions may be represented

by their value oracles or by total membership access to their graphs, with the usual effective equivalence. This equivalence does not extend unchanged to partial functions.

2.1 Mass problems and the direction of reduction

For classes $\mathcal{P}, \mathcal{Q}$ of total coded objects, Muchnik reducibility is

$$\mathcal{P} \leq_w \mathcal{Q} \iff \forall q \in \mathcal{Q} \exists p \in \mathcal{P} (p \leq_T q).$$

The class on the right supplies the input. Medvedev reducibility $\mathcal{P} \leq_s \mathcal{Q}$ requires a *single* oracle functional producing an element of $\mathcal{P}$ from every element of $\mathcal{Q}$. A functional need only satisfy its output specification on valid inputs.[8, 4]

Two spectra must not be confused:

$$\begin{aligned} \text{Spec}_T(C) &= \{\deg_T(x) : x \in C\}, \\ \text{Up}(\mathcal{P}) &= \{\mathbf{d} : \exists p \in \mathcal{P} (\deg_T(p) \leq \mathbf{d})\}. \end{aligned} \tag{1}$$

The first records degrees of actual representatives of a class. The second records degrees *capable of computing* a solution to a mass problem. Report B's upward-spectrum notation is standardized here to Up; Report C's literal representative spectrum remains Spec_T .

Proposition 2.1 (Reducibility and computing spectra). $\mathcal{P} \leq_w \mathcal{Q}$ if and only if $\text{Up}(\mathcal{Q}) \subseteq \text{Up}(\mathcal{P})$.

Proof. If a degree computes a member of $\mathcal{Q}$, the reduction supplies a member of $\mathcal{P}$ below it, proving inclusion. Conversely, for $q \in \mathcal{Q}$, the degree of q belongs to $\text{Up}(\mathcal{Q})$ and hence to $\text{Up}(\mathcal{P})$; some $p \in \mathcal{P}$ therefore satisfies $p \leq_T q$. No uniform selection of these witnesses is needed. $\square$

2.2 The formal objects must match the classical statement

The inspected Mathlib degree quotient is built from partial functions. The original audits warn against simply substituting it for the quotient of total set oracles. Report A identifies a separate `SetTuringDegree` development in ProveIt. Section 13 preserves both that constructive evidence and the limits of the audits. None of the mathematical questions in this report is reinterpreted as a claim about arbitrary partial-function degrees.[9, 10, 11]

3 Consolidated catalogue

The identifiers below are stable throughout the report and the CSV crosswalk. Evidence codes describe the source record, not an exhaustive certification of present open status. The V entries are specifically subject to the convention and proof-validation qualifications in Section 4. A split H/E rating means a high-dependency whole target with an exploratory restricted version.

ID	Topic	Tier	Evidence
Towers			
T1	Weak set tower to strict set tower	V	O26 / V
T2	Function tower to strict set tower	V	O26 / V
T3	Weak set tower to function tower	V	O26 / V
T4	Function tower to weak set tower	V	O26 / V
Coarse and effective-dense classes			
C1	Least Turing degree in each nonuniform coarse class	B	O25
C2	Least Turing degree in each nonuniform effective-dense class	E	O25

ID	Topic	Tier	Evidence
C3	Binary effective-dense representative below the jump	E	O25 / Ref
Metric geometry			
M1	Five-cycle realization in the degree metric	B	O21
M2	Finite and countable graph-metric universality	H	O21
M3	Distance-one-half graph: diameter three or four	B	O21
M4	Random-degree finite extension property	H	O21
M5	Random incomparable partner at distance one	H	O21
M6	Hyperimmune-free versus hyperimmune PA degrees	H	O21
M7	Does dispersiveness imply AEWG?	H	O26
Local oracle classes			
Q1	Classify ceer-diagonalizing degrees; is Diag = PA?	E	O25 / Ref
Q2	Do nonzero index-guessable degrees contain 1-generics?	E	O25
Q3	Does computing no tower imply index guessability?	E	O25 / O26
Lattices and local order			
L1	Strong minimal covers of minimal degrees	H	S07
L2	Decide finite-lattice embeddability in c.e. degrees	H	O21 / Ref
L3	Finite lattice detecting beyond-omega-squared fickleness	H	O21
L4	Meet-removal invariance of bounding spectra	H	O21
L5	Join-semidistributive final segments of d.c.e. degrees	H	O24
Categoricity			
K1	Categoricity versus decidable categoricity	H/E	O25
K2	Infinite topologically computably categorical Polish space	H	O25
K3	Categoricity degree of Baire space	H	O25
K4	Realizing c.e. topological categoricity degrees	H/E	O25 / Ref
Randomness			
N1	Comparable 2-random reals with equal jumps	H	O26
N2	Comparable 2-random reals with bounded complexity difference	H	O26
Dimension			
D1	Classical Hausdorff dimension of minimal-degree reals	H	O25
D2	Effective dimension one for the minimal-degree class	H	O25
D3	Classical Hausdorff dimension of maximal antichains	H	O25
D4	Relative effective dimension of maximal antichains	H	O25
Global structure			
G1	Rigidity of all Turing degrees	H	O26
G2	Rigidity below the halting degree	H	S17
G3	Martin's conjecture, Part I	H	O26
G4	Martin's conjecture, Part II	H	O26
G5	Steel's uniformity conjecture	H	S12
G6	Universality of Turing equivalence	H	S12
G7	Sacks's order-embedding conjecture	H	O26

R1–R3, discussed in Section 12, are separately proposed programmes: bounded lattice-obstruction classification, a new restricted meet-removal theorem, and finite bookkeeping for measure constructions. Their specification and novelty remain part of the research task.

4 Tower comparisons: proposed proofs and their exact scope

This section integrates Report B's proposed affirmative resolutions of T1–T4 with Report C's diagnostic objections to simpler codings.[2, 3] The main mathematical content is two explicit transformations and their inverse obstruction arguments. They use no priority construction, jump oracle, search for stabilization stages, or effective listing of computable indices.

4.1 Precise representations and a necessary convention check

A sequence of sets (G_n) is presented by the single oracle $\{\langle x, n \rangle : x \in G_n\}$, for a fixed computable pairing bijection with computable inverse. A function sequence is presented by the total function $(n, x) \mapsto f_n(x)$, or equivalently by its coded graph. Each column is required to be computable, but the whole array need not be computable. We do *not* assume an effective sequence of column indices.

Definition 4.1 (The three tower problems). The class $\mathcal{T}$ consists of sequences of infinite computable sets such that $G_{n+1} \subseteq^* G_n$, $G_n \setminus G_{n+1}$ is infinite, and no infinite computable R satisfies $R \subseteq^* G_n$ for every n .

The class $\mathcal{T}^-$ consists of sequences of infinite computable sets linearly ordered by $\subseteq^*$, with no such computable R . Their list order need not be descending, and strict decrease is not required.

The class $\mathcal{TF}$ consists of sequences of unbounded total computable functions such that $f_{n+1} \leq^* f_n$, $f_{n+1} <^\infty f_n$, and no unbounded computable h satisfies $h \leq^* f_n$ for all n .

These are the explicit computable-column conventions adopted here for interpreting the questions in [4], Definitions 1.13, 2.4, and 2.18, with one issue made explicit: Definition 2.18 says “sets” without repeating infinitude and computability. The surrounding framework and its strong-finite-intersection observation support the interpretation above, but the omitted words should be checked with the author before claiming to have settled the intended published question. Read literally without infinitude, the constant empty sequence would satisfy the displayed weak-tower condition and make it a different, degenerate mass problem. Our mathematical statements below concern exactly the definition given here.

Some conventions impose $G_0 = \mathbb{N}$, as in the original tower framework [8]. Both constructions below have infinite first complement, so prepending $\mathbb{N}$ yields that normalization without changing maximality or computability. We prove this explicitly rather than assuming that arbitrary towers admit it.

4.2 Finite intersections: normalization without strictness

Lemma 4.2 (Intersection normalization). *Let (G_i) have infinite columns linearly ordered by $\subseteq^*$, and put $I_n = \bigcap_{i \leq n} G_i$. Each I_n is infinite, $I_{n+1} \subseteq I_n$, and (I_n) has a computable pseudointersection exactly when (G_i) does. If the original columns are computable, so are the I_n , and the intersection array is uniformly computable from the original array.*

Proof. In a finite list linearly ordered modulo finite sets there is a member G_j below all others in the preorder. Then $G_j \setminus I_n$ is contained in the finite union $\bigcup_{i \leq n} (G_j \setminus G_i)$. Removing this finite set leaves an infinite subset of I_n . The displayed exact containment follows directly from the definition. If $R \subseteq^* G_i$ for every i , then $R \setminus I_n$ is a finite union of finite sets. Conversely, $R \subseteq^* I_i \subseteq G_i$ gives $R \subseteq^* G_i$.

Membership in I_n uses the bounded conjunction of the first $n + 1$ column queries; this gives the uniform array operation and individual column computability. The existence of a least

member of the finite preorder was used only to prove infinitude. The algorithm does not attempt to select its index or recognize almost containment. $\square$

Nothing here guarantees an infinite difference $I_n \setminus I_{n+1}$. Report C correctly identifies this as a failure of finite intersections *alone*. The next construction adds a removable infinite layer and a bounded projection; it does not try to recognize an infinite difference effectively.

4.3 T1. Weak towers uniformly produce strict towers

Theorem 4.3 (Proposed weak-to-strict reduction). *Under the stated conventions, $\mathcal{T} \leq_s \mathcal{T}^-$.*

Proof. Given $(G_n) \in \mathcal{T}^-$, put

$$I_n = \bigcap_{i \leq n} G_i, \quad H_n = \{\langle x, y \rangle : x \in I_n \text{ and } n \leq y \leq x\}. \quad (2)$$

Each H_n is computable, and the entire sequence is uniformly computable from the input array. This uses only finitely many membership queries.

There is exact containment $H_{n+1} \subseteq H_n$. By Lemma 4.2, infinitely many $x \in I_n$ satisfy $x \geq n$. The pairs $\langle x, n \rangle$ for these x belong to $H_n \setminus H_{n+1}$. This gives both infinitude and an infinite strict difference at every step.

Suppose, for a contradiction, that an infinite computable P satisfies $P \subseteq^* H_n$ for every n . Define

$$R = \{x : (\exists y \leq x) \langle x, y \rangle \in P\}. \quad (3)$$

This is a computable set because the existential search is bounded. All but finitely many members of P lie in H_0 , hence in the triangle $\{\langle x, y \rangle : y \leq x\}$. Within that triangle each first-coordinate fiber is finite. Therefore R is infinite: finitely many first coordinates could account for only finitely many of those members of P .

For fixed n , if $x \in R \setminus I_n$, choose $y \leq x$ with $\langle x, y \rangle \in P$. Since $x \notin I_n$, this pair is not in H_n . Consequently,

$$R \setminus I_n \subseteq \pi_1(P \setminus H_n), \quad (4)$$

where π_1 is the first-coordinate projection through the pairing inverse. The right side is finite. Thus $R \subseteq^* I_n \subseteq G_n$ for every n , contradicting the input's maximality.

Finally, $\mathbb{N} \setminus H_0$ is infinite, since all pairs with $y > x$ are excluded. When a first column equal to $\mathbb{N}$ is required, output $\mathbb{N}, H_0, H_1, \dots$. The first difference is infinite and all remaining conditions have already been proved. The same fixed algorithm works for every valid input, so the reduction is Medvedev, not merely Muchnik. $\square$

Why the padding is doing real work. Taking finite intersections alone gives a descending sequence but may repeat modulo finite sets. The extra coordinate y supplies an infinite layer to remove at each stage. Allowing arbitrary y would make projection of a computable pseudointersection only visibly c.e.; the bound $y \leq x$ repairs that problem. It simultaneously ensures finite fibers and decidable projection. This is the central design constraint, not a cosmetic coding choice.

4.4 Diagnostic: unpadded subgraphs leave a computable row

Proposition 4.4 (Why the unpadded subgraph coding fails). *Given any $g \in \mathcal{TF}$, put $\hat{g}_e(n) = g_e(n) + 1$. Then $\hat{g} \in \mathcal{TF}$. But the subgraph sets*

$$S_e = \{\langle n, k \rangle : k < \hat{g}_e(n)\}$$

have the computable infinite common subset $R = \{\langle n, 0 \rangle : n \in \mathbb{N}\}$, so they are not a maximal set tower, weak or strict.

Proof. Adding one preserves unboundedness, eventual decrease and infinitely many strict decreases. For any computable unbounded h , the function $q(n) = \max(h(n) - 1, 0)$ is computable and unbounded. For some e there are infinitely many n with $g_e(n) < q(n)$. At those inputs, $g_e(n) + 1 < h(n)$. Thus $\hat{g}$ retains maximality. Every $\hat{g}_e(n)$ is positive, so every $\langle n, 0 \rangle$ belongs to every S_e . $\square$

The proposition rejects one natural coding, not the existence of all oracle-computable conversions. An improved coding must prevent this bounded-height escape.

4.5 T2. Function towers uniformly produce set towers

Theorem 4.5 (Proposed function-to-set reduction). *Under the stated conventions, $\mathcal{T} \leq_s \mathcal{TF}$.*

Proof. For $(f_n) \in \mathcal{TF}$, define

$$H_n = \{\langle x, y \rangle : n \leq y < f_n(x)\}. \quad (5)$$

Membership requires a single value of the input array. Every column is computable. Since f_n is unbounded, infinitely many x satisfy $f_n(x) > n$: otherwise the finitely many exceptional values and the bound n would bound its whole range. Hence the layer $\{\langle x, n \rangle : f_n(x) > n\}$ is infinite. It is contained in $H_n \setminus H_{n+1}$.

Let $E_n = \{x : f_{n+1}(x) > f_n(x)\}$, a finite set. Outside E_n , membership in H_{n+1} implies membership in H_n . Above each exceptional x , only finitely many $y < f_{n+1}(x)$ occur. Thus

$$H_{n+1} \setminus H_n \subseteq \{\langle x, y \rangle : x \in E_n, y < f_{n+1}(x)\}$$

is finite. The required almost containment follows even though the functions need not be pointwise decreasing.

Suppose P is an infinite computable pseudointersection of all H_n . Define the total computable function

$$h(x) = \max(\{0\} \cup \{y + 1 : y < f_0(x) \text{ and } \langle x, y \rangle \in P\}). \quad (6)$$

The maximum is over an explicitly bounded finite set. Computability here uses P and the single ordinary computable function f_0 ; there is no demand to extract an index for f_0 uniformly from the original array.

For every k , the set $P \cap H_0 \cap H_k$ is infinite, since its complement in P is finite. Any pair $\langle x, y \rangle$ in it has $y \geq k$ and $y < f_0(x)$, so $h(x) \geq y + 1 > k$. Hence h is unbounded.

For fixed n , suppose $h(x) > f_n(x)$. The maximum in (6) is then positive and attained at $y + 1$ for some $\langle x, y \rangle \in P$. The integer inequality $y + 1 > f_n(x)$ implies $y \geq f_n(x)$, so the pair is not in H_n . It follows that

$$\{x : h(x) > f_n(x)\} \subseteq \pi_1(P \setminus H_n). \quad (7)$$

The right side is finite, and therefore $h \leq^* f_n$ for every n . This contradicts maximality of the function tower.

For the optional normalization, $\mathbb{N} \setminus H_0$ is infinite because $\langle x, f_0(x) \rangle \notin H_0$ for every x . Prepend $\mathbb{N}$ as before. All output queries are uniform, proving the asserted strong reduction. $\square$

Uniformity of the forward map versus its correctness proof. The forward transformation (5) uses no ordinary index for any f_n : it queries their combined oracle. In the contradiction argument, computability of h is an existential assertion about an ordinary program. A program for the one fixed computable f_0 may be used to prove that assertion. Requiring a uniform algorithm that recovers such an index from the array would strengthen the argument unnecessarily and could destroy it. This distinction should be visible in the Lean theorem types.

4.6 T3–T4. The reverse conversion and the combined conclusion

For completeness, the easy reverse conversion is included. Its conclusion is already Proposition 2.21 of [4]; it is not claimed as new.

Lemma 4.6 (Set towers to function towers). $\mathcal{TF} \leq_s \mathcal{T}$.

Proof. For a strict set tower (G_n) , put

$$f_n(x) = \begin{cases} x, & x \in G_n, \\ 0, & x \notin G_n. \end{cases}$$

Infinitude of G_n makes f_n unbounded. Almost containment of successive sets gives eventual domination; the infinite set $G_n \setminus G_{n+1}$, with 0 removed, gives infinitely many strict decreases.

If h were an unbounded computable eventual lower bound for every f_n , its support $R = \{x : h(x) > 0\}$ would be computable and infinite. Outside the finite exceptional set where $h(x) > f_n(x)$, a point of R must belong to G_n . Thus $R \subseteq^* G_n$ for every n , a contradiction. The conversion is uniform on the combined input oracle. $\square$

Corollary 4.7 (Proposed resolution of the four comparisons). *For the definitions in this section,*

$$\boxed{\mathcal{T} \equiv_s \mathcal{T}^- \equiv_s \mathcal{TF}.} \tag{8}$$

In particular, all four Muchnik comparisons T1–T4 hold, and their upward-closed Turing-degree spectra coincide.

Proof. Theorem 4.3 gives $\mathcal{T} \leq_s \mathcal{T}^-$, while forgetting strict list order gives $\mathcal{T}^- \leq_s \mathcal{T}$. Theorem 4.5 and Lemma 4.6 give $\mathcal{T} \equiv_s \mathcal{TF}$. Transitivity proves $\mathcal{TF} \leq_s \mathcal{T}^-$ and $\mathcal{T}^- \leq_s \mathcal{TF}$. Strong reducibility implies weak reducibility, and (1) gives the spectrum statement. $\square$

Status of this corollary. The supplied proof proposal is the basis for tier V, not a claim of external acceptance. It has not been checked in Lean. No independent expert review of either the arguments or the interpretation of the source is recorded in the supplied reports. Novelty also remains unverified: elementary padding arguments may exist under other terminology. Even if the source’s intended weak-tower convention differs, Theorems 4.3 and 4.5 remain sharply stated claims with proofs to check; their advertised relationship to the published questions would then have to be revised. The function-to-set direction has less definitional ambiguity than the weak-tower directions.

A useful strengthening, not an extra claimed open problem. The proof of Theorem 4.3 uses linear order only to obtain infinite finite intersections. It therefore applies to any sequence of computable sets with that property and with no infinite computable pseudointersection. This observation is an immediate consequence of the proof. It is a useful formal abstraction and a useful target for a literature search; it is not counted as another independent discovery.

5 Coarse and effective-dense representative spectra

5.1 Descriptions, reductions, and least representatives

For $S \subseteq \mathbb{N}$ put $\rho_n(S) = |S \cap [0, n]|/n$ for $n > 0$. Define

$$\text{CD}(f) = \{h \in \mathbb{N}^{\mathbb{N}} : \rho_n(\{k : h(k) \neq f(k)\}) \longrightarrow 0\}.$$

An effective-dense description of f is a *total* function $h : \mathbb{N} \rightarrow \mathbb{N} \cup \{\perp\}$ that never gives an incorrect numerical answer and has a density-zero set of $\perp$ answers. Write $\text{ED}(f)$ for this class. The undefined-answer symbol $\perp$ is an explicit output value, not a diverging computation. For $r = \text{nc}$ or ned define

$$f \leq_r g \iff \forall h \in D_r(g) \exists k \in D_r(f) (k \leq_T h), \quad D_{\text{nc}} = \text{CD}, \quad D_{\text{ned}} = \text{ED}.$$

These are nonuniform coarse and nonuniform effective-dense reducibility. Equivalence means reducibility in both directions [5].

5.2 C1. Must every nonuniform coarse class have a least Turing degree?

Assessment: B, counterexample route. The coarse instance of Gerdes’s Question 7 asks whether

$$\forall f \in \mathbb{N}^{\mathbb{N}} \exists g \equiv_{\text{nc}} f \forall h \equiv_{\text{nc}} f (g \leq_T h).$$

A small negative example would settle this instance completely. The distinction between a *minimal* element and a *least* element must be preserved: a least representative is below every representative.

A concrete sufficient target is a nonzero coarse class with two representatives whose Turing degrees form a minimal pair. Proposition 5.2 proves why this suffices. The research difficulty is arranging those properties together. Ordinary Turing noncomputability is far too weak: Proposition 5.1 gives every Turing degree a representative in the computable coarse class. The **B** rating is conditional on obtaining a short finite-extension argument, not on importing a large minimal-pair theorem as an unproved Lean axiom. Unlike the tower validation track, this remains a construction-search target; its formal cost depends strongly on the availability of an oracle-construction library.

5.3 C2. Least Turing degrees in effective-dense classes

Assessment: E. Take the preceding least-representative statement with $\equiv_{\text{ned}}$ in place of $\equiv_{\text{nc}}$; this is another instance of [5, Question 7]. Uniform variants are distinct questions and are not implicitly settled by either nonuniform instance.

The minimal-pair criterion still applies because it is an abstract fact about equivalence classes. What does not transfer automatically is the density-zero modification argument: an effective-dense description cannot output a wrong value at *any* location. A description accurate for one function may therefore fail to describe a density-zero modification. An AI proof that treats wrong answers and explicit omissions interchangeably has changed the problem. This makes a shared formal development useful, but makes C2 a less immediate short target than C1.

5.4 C3. How low can a set representative be?

Assessment: E. A precise component of Gerdes’s Question 5 is

$$(\exists X \in 2^{\mathbb{N}} f \equiv_{\text{ned}} X) \implies (\exists Y \in 2^{\mathbb{N}} [f \equiv_{\text{ned}} Y \wedge Y \leq_T f']).$$

The source also asks for arithmetical or hyperarithmetical bounds [5]. The displayed statement isolates the $\Delta_2^0(f)$ version.

A disproof of this bound would answer that component, not the weaker hyperarithmetical question. A proof with f'' in place of f' would be an explicitly weaker result. For a short project, the appealing branch is a bounded, finite-search selection argument. The main risk is that selecting among set representatives involves a much stronger basis theorem than the displayed formula suggests. Hyperarithmetical versions additionally require ordinal and effective descriptive-set-theoretic infrastructure. Keep those costs separate from the f' target.

5.5 Diagnostic lemmas: what a counterexample must establish

Proposition 5.1 (Sparse coding and coarse Turing spectra). *For binary sequences, the Turing-degree spectrum of a nonuniform coarse-equivalence class is upward closed. The spectrum of the computable coarse class is all of $\mathcal{D}$.*

Proof. Let X represent a given coarse class and let B have degree at least $\deg_T(X)$. Define Y by $Y(2^n) = B(n)$ and $Y(k) = X(k)$ elsewhere. The powers of two have density zero, so X and Y have exactly the same coarse descriptions: a union of two density-zero error sets still has density zero. Hence they are uniformly, and therefore nonuniformly, coarse equivalent. Also $B \leq_T Y$ by reading powers of two, while $Y \leq_T B$ because $X \leq_T B$. Thus $\deg_T(Y) = \deg_T(B)$, proving upward closure. Take X to be constantly zero to obtain every degree in the computable coarse class. That class also contains a computable representative, so its least Turing degree is $\mathbf{0}$. $\square$

This is a useful warning for C1: even the presence of arbitrarily complicated representatives says nothing against the existence of a least one. The same proof of upward closure works for natural-number-valued representatives, with a fixed encoding of the higher oracle.

Proposition 5.2 (A sufficient certificate for no least degree). *Let E be an equivalence relation on total functions. Suppose an E -class C contains no computable member and has $f, g \in C$ such that*

$$h \leq_T f \wedge h \leq_T g \implies h \text{ is computable}$$

for every total function h . Then $\text{Spec}_T(C)$ has no least element.

Proof. A representative $h \in C$ of a least degree would satisfy $h \leq_T f$ and $h \leq_T g$. It would be computable by the assumption, contradicting the first property of C . $\square$

This gives a sufficient route for C1 or C2, not a necessary characterization. In particular, a proof that no such minimal pair exists would not prove that every class has a least degree.

6 Metric questions: the most bounded full-target bets

6.1 The metric used in these questions

For $A, B \subseteq \mathbb{N}$, define the upper-density discrepancy

$$\delta(A, B) = \limsup_{n \rightarrow \infty} \frac{|(A \triangle B) \cap [0, n]|}{n}, \quad n \geq 1.$$

Quotienting by $\delta = 0$ gives a metric space $\mathcal{S}$. For $\mathbf{a} \in \mathcal{D}$, let $C_{\mathbf{a}}$ be the closure in $\mathcal{S}$ of the coarse classes of its representatives. Put

$$H(\mathbf{a}, \mathbf{b}) = \max \left\{ \sup_{x \in C_{\mathbf{a}}} \inf_{y \in C_{\mathbf{b}}} \delta(x, y), \sup_{y \in C_{\mathbf{b}}} \inf_{x \in C_{\mathbf{a}}} \delta(x, y) \right\}. \quad (9)$$

Hirschfeldt–Jockusch–Schupp establish that H is a degree metric taking only the values $0, 1/2, 1$. [12] The distance-one graph has vertices $\mathcal{D}$ and edges $H = 1$; its complement on distinct vertices has edges $H = 1/2$. The word “metric” here is not a computable presentation of all degrees.

6.2 M1. A five-cycle realization B

Find distinct $\mathbf{a}_0, \dots, \mathbf{a}_4$ satisfying, with indices modulo five,

$$H(\mathbf{a}_i, \mathbf{a}_j) = \begin{cases} 0 & i = j, \\ 1 & i - j \equiv \pm 1 \pmod{5}, \\ 1/2 & i - j \equiv \pm 2 \pmod{5}. \end{cases} \quad (10)$$

Alternatively, prove that no such tuple exists.

Source and status. HJS explicitly single out this test in Question 9.3. The reviewed open-question text is from 2021; the paper appeared in 2024. Report A located no later resolution in its 17 September audit. [12]

Why this deserves a bounded attempt. There is one finite configuration and only ten unordered pair constraints. A counterexample to general metric universality would already be obtained by proving nonrealizability. A positive solution need not classify arbitrary finite graphs or provide an extension theorem over an arbitrary previously chosen tuple. Thus the existential target can be separated from much stronger uniform construction goals.

That separation is especially useful for AI-assisted proof search. One can keep a fixed list of obligations for each pair, ask for candidate constructions, and reject a candidate as soon as one lower or upper bound fails. The relevant search objects should be expressions in degree-building operations together with *proved* distance lemmas, not guessed numerical distances for finitely simulated oracles.

A useful obstruction to a naive strategy. HJS realize comparability graphs as distance-one configurations. [12] The following elementary observation shows why that theorem alone does not answer M1.

Proposition 6.1. *A triangle-free comparability graph is bipartite. In particular, C_5 is not a comparability graph.*

Proof. Orient each edge according to the partial order. If a vertex has an incoming and an outgoing edge, $u < v < w$, transitivity supplies the edge uw , producing a triangle. Every nonisolated vertex therefore has only incoming edges or only outgoing edges. These two kinds of vertices form a bipartition. An odd cycle cannot be bipartite. $\square$

This is a known-type finite graph observation, not a new theorem about degrees. It rules out one proof template, not a Turing-degree realization. Likewise, every assignment of distances $1/2$ and 1 to distinct points satisfies the triangle inequality: its largest distance is at most the sum of any two nonzero distances. Metric consistency is not the obstruction. The archive includes a small exhaustive finite check of these observations, explicitly segregated from the open problem.

Concrete attempt design. Start with three candidate templates: finite joins of a small family of mutually relative random reals; extensions of one genericity construction; and a direct priority construction with ten pairwise targets. For each template, write the *actual* oracle and density

lemmas that would determine every entry of the matrix. Do not assume that independence, nonreducibility, or randomness automatically gives the desired Hausdorff distance. Discard a template that forces a comparability graph, a constant off-diagonal distance, or an unproved universal extension principle.

A negative route is also worth testing. Seek a universal five-point implication of the form

$$\bigwedge_{i < 5} H(\mathbf{a}_i, \mathbf{a}_{i+1}) = 1 \implies \bigvee_{i < 5} H(\mathbf{a}_i, \mathbf{a}_{i+2}) \neq \frac{1}{2}.$$

The statement has small syntactic scope even though proving it might require substantial computability theory. A short structural obstruction would likely be easier to formalize than a fresh random-real construction.

Paper-and-Lean assessment. This is the strongest **B** candidate in the survey. Its weeks-scale case depends on finding a proof whose unformalized prerequisites stay small. A solution that invokes several deep randomness, basis, and priority theorems would lose this advantage. The inherited code audit does not establish that those prerequisites are already available.

6.3 M2. Finite and countable metric universality H

For every finite graph F , does there exist an injective map $e : V(F) \rightarrow \mathcal{D}$ such that, for distinct u, v ,

$$H(e(u), e(v)) = 1 \iff \{u, v\} \in E(F), \quad H(e(u), e(v)) = 1/2 \iff \{u, v\} \notin E(F)?$$

Ask the same question for every countable graph. These are the finite and countable versions of HJS Question 9.3.[12]

The countable version is not obtained merely by solving each finite instance: independently chosen finite witnesses need not extend one another. A compactness argument would require a justified bridge from an abstract model or limit object back to actual Turing degrees. M1 is a genuine instance of the finite question, not an equivalent reformulation.

Paper-and-Lean assessment. A uniform construction would need an extensible invariant governing arbitrarily many pair requirements. That is a much larger formal target than ten constraints. Prefer a finite obstruction or a new named family of realizable graphs before attempting universality.

6.4 M3. Diameter three or four? B

Let K be the graph on $\mathcal{D}$ with adjacency $H(\mathbf{a}, \mathbf{b}) = 1/2$. HJS show that its diameter belongs to $\{3, 4\}$ and ask which value occurs.[12, Question 9.7]

There are two exact proof targets. Define a reflexive relation

$$E(\mathbf{a}, \mathbf{b}) \iff \mathbf{a} = \mathbf{b} \vee H(\mathbf{a}, \mathbf{b}) = 1/2.$$

An upper bound of three is equivalent to

$$\forall \mathbf{a}, \mathbf{b} \exists \mathbf{c}, \mathbf{d} \quad E(\mathbf{a}, \mathbf{c}) \wedge E(\mathbf{c}, \mathbf{d}) \wedge E(\mathbf{d}, \mathbf{b}). \quad (11)$$

A lower bound of four is obtained by exhibiting $\mathbf{a}, \mathbf{b}$ for which the displayed existential statement fails. Equality is included in E so that paths of length less than three are handled correctly.

Why a short argument might exist. The question is a one-step improvement of an existing bound, not an unconstrained classification problem. A promising search is to inspect the four-edge construction and identify whether its two middle vertices can be merged or replaced by a single uniformly suitable degree. The dual search is for two endpoint properties that make every three-edge path impossible. Both are coherent local proof-development tasks.

A graph solver can test whether a proposed list of pairwise constraints logically allows a short path. It cannot show that an abstract graph it constructs is the actual half-distance graph, nor quantify over all degrees. The tool should therefore verify deductions from established distance lemmas, not manufacture those lemmas.

Paper-and-Lean assessment. Marked **B** as a second, more conditional bet. The exact-diameter theorem also needs the already known complementary bound in Lean. Proving only (11) under an assumed four-edge theorem is not an end-to-end formalization of the published conclusion. If the necessary HJS infrastructure dominates the project, downgrade this to a longer programme immediately.

6.5 M4. Random-degree extension property H

Given disjoint finite sets A, B of degrees containing Martin–Löf random reals, must there be a random degree $\mathbf{c}$ satisfying

$$(\forall \mathbf{a} \in A) H(\mathbf{a}, \mathbf{c}) = 1/2, \quad (\forall \mathbf{b} \in B) H(\mathbf{b}, \mathbf{c}) = 1?$$

This is HJS Question 9.4. Here “random degree” means *contains* a random representative.[12]

For nonempty $A \cup B$, the distance requirements themselves force freshness. Repeated use of a positive theorem of this form would construct arbitrary countable graph patterns inside the random degrees. The extension property is therefore considerably stronger than solving an isolated finite configuration. The analogous unrestricted extension property is already false; removing the word “random” is not a helpful weakening.[12]

Paper-and-Lean assessment. The likely proof obligations combine relative randomness, avoidance of finitely many degree constraints, and exact density estimates. A project restricted to one carefully chosen nontrivial pattern could be smaller, but its open status must be checked rather than inferred from the general question.

6.6 M5. A random incomparable partner at distance one H

For every random degree $\mathbf{a}$, is there a random degree $\mathbf{b}$ with

$$\mathbf{a} \not\leq \mathbf{b}, \quad \mathbf{b} \not\leq \mathbf{a}, \quad H(\mathbf{a}, \mathbf{b}) = 1?$$

This is HJS Question 9.5.[12]

The universal quantifier over the starting degree is an important source of difficulty. Producing one pair, or a partner only for degrees satisfying an additional lowness condition, would not settle this statement. Conversely, a verified obstruction for one random starting degree would refute it.

Paper-and-Lean assessment. This is narrower than M4 but still lacks an identified dependency-light route. A useful preliminary exercise is to separate the proof of incomparability from the proof of distance one and determine whether the two constructions can be combined without changing the prescribed starting degree.

6.7 M6. Hyperimmune-free and PA degrees H

Must

$$H(\mathbf{a}, \mathbf{b}) = 1$$

hold whenever $\mathbf{a}$ is hyperimmune-free and $\mathbf{b}$ is both hyperimmune and a PA degree? This is HJS Question 9.6.[12]

Hyperimmune-free means that every total function computable from the degree is dominated by a computable function; hyperimmune means not hyperimmune-free. A PA degree computes a complete consistent extension of Peano arithmetic. These conditions are substantive oracle properties, not labels on vertices of a finite graph.

Paper-and-Lean assessment. The form of the statement invites a contradiction proof extracting a forbidden domination or approximation property from $H < 1$. That is a plausible mathematical angle, but formalizing the relevant PA-degree and domination machinery would be a substantial additional task. It is not ranked above M1 merely because its displayed formula is shorter.

6.8 M7. Dispersive versus AEWG H

A degree $\mathbf{a}$ is *dispersive* if $H(\mathbf{a}, \deg X) = 1$ for almost every $X \in 2^{\mathbb{N}}$. It is *attractive* if the almost-everywhere distance is $1/2$. A set A is *AEWG* when almost every oracle computes a set weakly 1-generic relative to A . Such a generic meets every dense A -c.e. family of finite binary strings.

The open question is

$$\text{dispersive}(\mathbf{a}) \implies \text{AEWG}(\mathbf{a})?$$

Its restriction to $\mathbf{a} \in \mathcal{R}$ remains open in Zhao's 16 September 2026 preprint, which also proves AEWG for every low₂ c.e. degree and constructs a superhigh c.e. example. AEWG already implies dispersiveness. Here low₂ means $\mathbf{a}'' = \mathbf{0}''$, and superhigh means $\mathbf{0}'' \leq_{\text{tt}} \mathbf{a}'$. [6]

Paper-and-Lean assessment. This has the freshest explicit open-status evidence in the inherited catalogue, but freshness is not tractability. A counterexample must defeat the almost-everywhere generic-computation property while preserving dispersiveness. A positive proof must cover cases beyond the low₂ theorem; one superhigh example does not establish the superhigh case. A new finite bookkeeping lemma extracted from these constructions is a more controlled **R** project, described in Section 12.

7 Focused local degree-class questions

7.1 Q1. Diagonalizing ceers: is Diag = PA?

A ceer is an equivalence relation on $\mathbb{N}$ with c.e. graph. Define Diag by

$$\mathbf{d} \in \text{Diag} \iff \forall E \neq \text{Id}_1 \exists g \leq_T \mathbf{d} \forall x \neg E(x, g(x)),$$

where E ranges over ceers, g is total, and Id_1 is the one-class relation. Let DNR be the degrees computing a total function h with $h(e) \neq \varphi_e(e)$ whenever the latter converges. Let PA be the binary-valued version of this degree class. The established bounds are $\text{PA} \subseteq \text{Diag} \subseteq \text{DNR}$ [13].

Question. Does $\text{Diag} \subseteq \text{PA}$ hold, or can one construct a non-PA degree in Diag? This is a focused sharpening of the open classification request in Problem 25(i) of [14]; the equality is a refinement, not a separately verified source-listed conjecture.

Assessment: E; the best unproved local classification target here. The statement is elementary enough to expose all quantifiers early. A positive route would extract binary

diagonalization power from the ability to avoid all ceer classes. A negative route would construct one oracle meeting the avoidance demands while preventing it from computing any binary DNR function. These are research directions, not established reductions.

The decisive pitfall is uniformity. The definition says $\forall E \exists g$, not that an index for g must be computable uniformly from an arbitrary index for E . A positive proof cannot quietly strengthen the hypothesis to such a uniform selector. A negative proof must deal with *every* eligible ceer, not just a finite menu. Merely coding a partition with two visible classes will not test the difficult cases. For Lean, ceer presentations and diagonal requirements are moderate infrastructure; a new infinite-injury construction would substantially enlarge the project.

7.2 Q2. Do nonzero index-guessable degrees contain 1-generics?

Fix standard enumerations of oracle functionals Φ_e and ordinary partial computable functions φ_e . An oracle L is index guessable when

$$\exists \Gamma \forall e [\Phi_e^L \text{ is total computable} \implies \Gamma^{\emptyset'}(e) \downarrow \wedge \varphi_{\Gamma^{\emptyset'}(e)} = \Phi_e^L]. \quad (12)$$

The guessing oracle is $\emptyset'$, not L or L' . No condition is imposed outside the promise in brackets [8].

A set $G \in 2^{\mathbb{N}}$ is 1-generic if for every c.e. set W of finite binary strings it either has an initial segment in W , or has an initial segment with no extension in W . The degree-level question is

$$\mathbf{d} > \mathbf{0} \wedge \text{IG}(\mathbf{d}) \implies \exists G \in \mathbf{d} [G \text{ is } \Delta_2^0 \text{ and 1-generic}] ?$$

It is a carefully scoped interpretation of the first reversal asked in Problem 23 [14]. The nonzero condition and “contains a representative” are essential: literal genericity of every equivalent set is not degree invariant, and the computable degree is an immediate obstruction to a naive statement.

Assessment: E. Begin by trying to turn the promise-based index guesser into a single generic representative, and identify exactly where meeting or avoiding a c.e. set of strings becomes effective. Alternatively, seek a nonzero index-guessable degree with an invariant excluding all generic representatives. Either approach has a relatively small statement but needs substantial recursion-theoretic machinery. Formalizing the known forward implication is a sensible calibration milestone, not new mathematics.

7.3 Q3. Does computing no maximal tower characterize index guessability?

Question. If no member of $\mathcal{T}$ is computable from L , must L satisfy (12)? The tower class is defined precisely in Section 4. This is the second reversal in Problem 23 and is explicitly asked by Lempp, Miller, Nies, and Soskova [8, 14].

The known forward implications run from Δ_2^0 1-genericity to index guessability, then to failure to compute a maximal tower, then to lowness. The last implication is not reversible [8]. Consequently, producing an arbitrary low oracle does not address Q3.

Assessment: E, with greater construction risk than the padding project. A direct positive proof must recover *indices* for arbitrary computable outputs, not just values of those outputs. A negative construction must simultaneously defeat every attempted tower code below its oracle and defeat every candidate guesser. A failure of uniform tower conversion is insufficient for a Muchnik separation: one needs a single source object from which *no* target object is computable.

An especially important limitation is that the positive comparisons proposed in Section 4 do not settle Q3. They show equality of several tower spectra under the stated conventions; they do not identify that common spectrum with the complement of the index-guessable degrees. The four V tower entries and Q3 therefore have quite different remaining mathematical content.

Reconciling the two genericity formulations. Report C phrases Q2 as containing a 1-generic representative, while Report B explicitly includes Δ_2^0 . Under the stated known implication chain, index-guessable degrees are low and hence below $\mathbf{0}'$; their set representatives are already Δ_2^0 . The displayed formulation records this scope rather than treating the two reports as asking incompatible questions. The broad classification of Diag and the equality $\text{Diag} = \text{PA}$ are not similarly identified: the latter remains an explicitly marked refinement.[8, 14]

8 Local structure, lattice spectra, and priority arguments

8.1 L1. Strong minimal covers of minimal degrees H

Call $\mathbf{a}$ minimal if

$$\mathbf{0} < \mathbf{a} \quad \text{and} \quad (\forall \mathbf{c} < \mathbf{a}) \mathbf{c} = \mathbf{0}.$$

A *strong minimal cover* of $\mathbf{a}$ is a degree $\mathbf{b}$ such that

$$\mathbf{a} < \mathbf{b} \quad \text{and} \quad (\forall \mathbf{c} < \mathbf{b}) \mathbf{c} \leq \mathbf{a}.$$

Does every minimal degree have a strong minimal cover?

Source and status. Lewis states this as the longstanding question of Yates, Question 1.1. The source inspected is from 2007; Report A located no subsequent resolution in its audit. Treat it as a historical open-problem lead requiring fresh specialist confirmation.[15]

The word “strong” is essential. Merely requiring no degree strictly between $\mathbf{a}$ and $\mathbf{b}$ allows degrees below $\mathbf{b}$ incomparable with $\mathbf{a}$; the displayed statement excludes them.

Paper-and-Lean assessment. Both positive and negative approaches must control every reduction below the newly constructed degree. Even a narrow special case may need a full tree construction. The existing Lean development does not supply minimal-degree existence, so a proof citing that theorem would already introduce a nontrivial formal dependency. A known special-case formalization is useful infrastructure but not a solution of this open problem.

8.2 L2. Finite lattices in the c.e. degrees

Question. Is there an algorithm deciding, from a finite lattice table, whether that lattice embeds into $\mathcal{R}$ as a lattice? Ko’s introduction records the unresolved finite-lattice embedding classification; this entry has older status evidence, not a newly confirmed 2026 resolution status [16].

Here an embedding is injective, preserves joins, and sends each meet to a *greatest lower bound among all c.e. degrees*. The image’s internal meet is not enough. Nor are the finite lattice’s bottom and top automatically required to map to $\mathbf{0}$ and $\mathbf{0}'$; imposing such endpoint conditions changes the problem. The decidability question is a precise component of the wider classification programme, rather than a promise of a particular structural characterization.

Assessment: H for the full problem. Finite algebra is attractive for AI search: one can enumerate tables, identify sublattices, and generate minimal combinatorial obstructions. But those operations do not establish realizability by c.e. degrees. A proposed obstruction needs a theorem about actual oracle computations; a proposed embedding needs sets, reductions, and the universal lower-bound property. A named, genuinely unresolved small lattice with a short realization criterion could become an E project after a fresh literature check. Merely solving its finite constraint table would not settle L2.

8.3 L3. A finite lattice detecting beyond- ω^2 fickleness H

For a finite lattice L , let its c.e.-degree bounding spectrum be

$$B(L) = \{\mathbf{d} \in \mathcal{R} : L \text{ has a lattice embedding into } [\mathbf{0}, \mathbf{d}]_{\mathcal{R}}\}.$$

Meets in this definition are greatest lower bounds among c.e. degrees. Use Ko's conventions on embeddings and designated bounds when translating a particular lattice presentation.

The target is a finite lattice L with

$$B(L) = \{\mathbf{d} \in \mathcal{R} : \mathbf{d} \text{ is not } \leq \omega^2\text{-fickle}\}.$$

Ko's paper studies this characterization problem and rules out several lattice shapes.[16]

What the ordinal measures. An ordinal-controlled approximation to $A \subseteq \mathbb{N}$ consists of a computable approximation $a(x, s)$ converging pointwise to its characteristic function, together with a computable descending rank assignment that must drop whenever the approximation changes. The definition uses a fixed effective ordinal-notation convention. For degrees, the bounded-fickleness assertion concerns *every representative*, not just one conveniently chosen c.e. set. The exact canonical-notation and initial-rank conventions should be copied from the source when formalizing; changing a strict bound to a non-strict one can change the target class.[16, Section 1.1]

Paper-and-Lean assessment. The lattice is finite, but the spectrum equality is two theorems quantified over infinitely many degrees: a construction below each degree satisfying the bound, and an extraction or obstruction argument for every embedding. The finite presentation does not make these proofs finite. The useful weeks-scale option is a new bounded obstruction classification, not an unsupported claim that checking small lattices solves the spectrum problem.

8.4 L4. Meet-removal invariance H

Ko's Conjecture 5.2 asks whether the paper's *meet-removal transformation* preserves the c.e.-degree bounding spectrum of the finite lattice configuration.[16] This is a precise source-indexed target: the transformed object is the enriched upper-semilattice presentation used there, not the lattice with some meet equations simply erased.

The transformation retains common-lower-bound restrictions and can require designated pairs to have no meet. For example, one kind of retained condition has the form

$$\forall \mathbf{w} \in \mathcal{R} \quad (\mathbf{w} \leq \mathbf{a} \wedge \mathbf{w} \leq \mathbf{b}) \implies \mathbf{w} \leq \mathbf{c},$$

where $\mathbf{w}$ is a degree variable. Such gates must be encoded before interpreting the conjecture as a finite syntactic transformation. A worked special case in Ko's Theorem 5.1 is already proved; it should not be rediscovered and presented as the conjecture's resolution.[16]

Paper-and-Lean assessment. There is a plausible short-project *restriction*: prove spectrum preservation for one new family under an explicit sufficient condition. The real challenge is to find a condition that is stronger than the existing examples but weaker than the full conjecture. Formalizing only a purely order-theoretic weakening implication is not enough if the converse needs an unformalized degree construction.

8.5 L5. Finite join-semidistributive final segments H

Does every finite join-semidistributive lattice occur as a final segment of the d.c.e. degrees? Such a lattice satisfies

$$x \vee y = x \vee z \implies x \vee (y \wedge z) = x \vee y.$$

The desired realization is an order isomorphism

$$L \cong \{\mathbf{e} \in \mathcal{D}_2 : \mathbf{d} \leq \mathbf{e} \leq \mathbf{0}'\} \quad \text{for some } \mathbf{d} \in \mathcal{D}_2.$$

Source and status. Lempp–Liu–Liu–Ng–Peng–Wu prove the finite distributive case and leave stronger realization questions, including the join-semidistributive case.[17]

Paper-and-Lean assessment. A final-segment theorem must exclude every additional d.c.e. degree above its bottom. Constructing only the required finitely many degrees proves an embedding, not an isomorphism onto the final segment. That exclusion obligation is the likely formal bottleneck. A named small nondistributive lattice would be a sensible instance only after checking that it is not already covered elsewhere; this survey does not label a guessed small instance as open.

9 Categoricity and algorithmic randomness

9.1 Computable presentations

A computable structure has an effective atomic diagram. It is $\mathbf{d}$ -categorical when every computable copy has an isomorphism to it computable in $\mathbf{d}$; a degree of categoricity is the *least* such degree. For decidable categoricity, the structure and copies have computable complete diagrams, allowing decisions of arbitrary first-order formulas. For topological categoricity, the objects are computable metric presentations of a common Polish space, and the maps are oracle-computable homeomorphisms of their completions. These are the conventions of [14].

9.2 K1. Categoricity versus decidable categoricity

Question. Is every degree of categoricity a degree of decidable categoricity? The realizing structure may change: this does not ask whether the original witness structure itself is decidable. The reverse inclusion is known; the cited problem specifically invites study of d.c.e. degrees [14], meaning degrees containing a difference of two c.e. sets.

Assessment: E for a precise d.c.e. subcase; H for an unrestricted classification project without a short proof. A successful restricted theorem must say whether it realizes a new degree, transfers an entire specified class, or proves an obstruction. Merely finding an isomorphism between two particular copies is not enough: categoricity quantifies over all allowed copies and requires leastness of the proposed degree: every sufficient oracle must compute it, not merely fail to be strictly below it. A useful AI-assisted first step is to translate one proposed coding construction into a small signature and list exactly which effective presentation operations it uses. Lean’s general model-theory vocabulary does not by itself discharge these effective coding obligations.

9.3 K2. An infinite topologically computably categorical space

Question. Does any infinite computable Polish space have topological categoricity degree $\mathbf{0}$ [14]?

Assessment: H end-to-end, despite the simple existential form. A familiar space in its usual presentation is not a solution: all computable metric presentations of the same topology must be considered. Two chosen presentations can look effectively identical while a third conceals noncomputable information in its local organization. A positive construction must supply an effective recovery method valid across this whole presentation class. A negative result must uniformly defeat every candidate infinite space, or otherwise prove a general obstruction. The formalization burden includes names for completion points and algorithms on those names, not merely classical topology.

9.4 K3. Does Baire space have a categoricity degree?

Question. Does $\mathbb{N}^{\mathbb{N}}$, as a topological space, have a least Turing degree computing homeomorphisms between all its computable presentations [14]?

Assessment: H. An upper bound on the complexity of homeomorphisms does not prove the existence of a least degree. Conversely, exhibiting one hard pair of copies does not show that no least degree exists. A project should separately state its uniform upper-bound theorem, its lower-bound coding theorem, and the argument that those bounds meet, or the argument that no minimum can exist. Formalizing only the standard product topology would miss the key issue.

9.5 K4. Which c.e. degrees are topological categoricity degrees?

Question. Is every c.e. degree realized in this way? The source uses the wording “every/any” [14]; we take the universal realization question as K4 and distinguish the weaker question whether some *nonzero* c.e. degree is realized.

Assessment: H for the universal claim; potentially E for a new, well-specified realization theorem with an existing effective-topology library. The nonzero qualification avoids a trivial finite-space interpretation of “any.” A construction must establish both that the proposed oracle suffices for all copies and that every sufficient oracle computes the intended degree. Those are different theorems. The formal endpoint should state the exact presentation model and what constitutes a computable homeomorphism, including its inverse where required by that convention.

9.6 Two fresh degree questions involving randomness

A 2-random real is Martin–Löf random relative to $\emptyset'$. One may formulate this through uniformly $\emptyset'$ -c.e. open tests (U_n) with measure at most 2^{-n} , requiring avoidance of $\bigcap_n U_n$ for every such test. Let $K(\sigma)$ be prefix-free Kolmogorov complexity for a fixed optimal prefix-free machine. The following two questions are explicitly Problem 1.4(1)–(2) in the August 31, 2026 version of [18]. They should not be confused with the weaker tasks obtained by dropping 2-randomness.

9.7 N1. Strictly comparable 2-randoms with equal jumps

Question. Are there 2-random x, z such that

$$x <_T z \quad \text{and} \quad x' \equiv_T z' ?$$

Assessment: H for the requested combined endpoint. This is an appealing, exact existence statement, and a targeted construction would be more manageable than a classification of all random degrees. Nevertheless, one must preserve 2-randomness, prove strict degree separation, and prove both jump reductions at once. A construction providing any two equal jumps or merely almost-everywhere randomness leaves the central requirements unresolved. Lean verification adds oracle tests, effective null sets, and a jump library to the dependency graph. A short conventional proof discovered with specialist insight could improve the rating; no such proof is supplied here.

9.8 N2. Strict degree increase with nearly identical complexity profiles

Question. Are there 2-random $x <_T z$ and a constant c such that

$$\forall n \quad |K(x \upharpoonright n) - K(z \upharpoonright n)| \leq c ?$$

Assessment: H. The condition is an absolute bounded difference, not $O(\log n)$ and not a one-sided inequality. Finite-prefix experiments cannot certify either randomness or the existence of one bound valid for all n . They may still expose where an encoding adds unwanted description overhead. The best first artifact is a precise coding lemma controlling this overhead, with an explicit decodable description transformation. Formalizing K and its invariance properties is additional work even when a good computability library is available. N1 and N2 are related source questions, not asserted to be equivalent.

10 Dimension of degree-invariant classes

These questions concern the usual metric on Cantor space, not the degree metric H of Section 6. A collection of degrees is represented by its *saturation*: all reals in those degrees, not a chosen transversal. Each individual Turing degree is countable and consequently has classical Hausdorff dimension zero; an uncountable union of such classes need not have dimension zero. The relevant sources distinguish Hausdorff from packing dimension and classical from effective dimension.[19]

For an oracle Z , let

$$\dim^Z(X) = \liminf_{n \rightarrow \infty} \frac{K^Z(X \upharpoonright n)}{n}, \quad \dim_{\text{eff}}^Z(C) = \sup_{X \in C} \dim^Z(X).$$

Here K^Z is prefix-free complexity relative to Z . We write $\dim_H(C)$ for ordinary Hausdorff dimension. A limsup complexity dimension is a different invariant. In particular, a packing-dimension theorem cannot simply be quoted as a Hausdorff-dimension theorem.[19]

10.1 D1. Classical Hausdorff dimension of minimal-degree reals

Let

$$\text{Min} = \{X \in 2^{\mathbb{N}} : \deg_T(X) > \mathbf{0} \text{ is a minimal degree}\}.$$

What is $\dim_H(\text{Min})$? Song–Yu explicitly records this question after Proposition 5.7.[19]

Paper-and-Lean assessment. H. A successful proof must control arbitrarily fine covers, not merely exhibit many minimal degrees or a class of packing dimension one. Even a short measure-theoretic argument could depend on a substantial minimal-degree construction. The inherited Lean audit does not supply that construction.

10.2 D2. Effective dimension one for the minimal-degree class

Does

$$\dim_{\text{eff}}^{\emptyset}(\text{Min}) = 1$$

hold? This is a distinct effective version of the minimal-degree dimension problem.[19] The supremum may equal one without being attained: proving the displayed statement is not automatically the same as producing one minimal-degree real of effective dimension exactly one. Conversely, one such real would suffice for the supremum statement.

Paper-and-Lean assessment. H. A useful candidate proof would provide, for each rational $q < 1$, a minimal-degree real whose lower complexity rate exceeds q . Its formalization must verify both minimality and the liminf lower bound. Replacing the liminf with a limsup changes the goal.

10.3 D3. Classical dimension of maximal antichains of nonzero degrees

Let $\mathcal{A} \subseteq \mathcal{D} \setminus \{\mathbf{0}\}$ be a maximal antichain in the nonzero degrees, and let $C_{\mathcal{A}}$ contain all reals in its degrees. Must

$$\dim_H(C_{\mathcal{A}}) = 1$$

hold? This is the classical Hausdorff form of the antichain question.[19] Excluding $\mathbf{0}$ is essential: otherwise the singleton consisting of the computable degree is a maximal antichain of all degrees, with a countable saturation of dimension zero.

Paper-and-Lean assessment. H. Maximality refers to comparability with every nonzero degree; it is not a maximality condition inside a conveniently chosen substructure. The saturation also matters: choosing one representative from each degree defines a different set and can alter dimension.

10.4 D4. Relative effective dimension of maximal antichains

For the same $C_{\mathcal{A}}$, must

$$\forall Z \in 2^{\mathbb{N}} \quad \dim_{\text{eff}}^Z(C_{\mathcal{A}}) = 1$$

hold? A particularly explicit test is whether a counterexample exists already for $Z = \emptyset'$. Song–Yu distinguish this relative question from the unrelativized effective dimension-one and classical packing dimension-one results already obtained.[19]

Paper-and-Lean assessment. H. An unrelativized lower bound cannot be relativized just by adding oracle superscripts. The proof must control the interaction between maximality in the ordinary degree order and the external oracle used in complexity. D3 and D4 are recorded separately so that a result for one is not advertised as a solution of the other.

11 Global structure and invariant functions

11.1 G1. Rigidity of all Turing degrees H

Is every order automorphism of $(\mathcal{D}, \leq)$ the identity?

$$\forall \pi \in \text{Aut}(\mathcal{D}, \leq) \quad \forall \mathbf{a} \in \mathcal{D} \quad \pi(\mathbf{a}) = \mathbf{a}.$$

Source and status. The automorphism/definability programme is discussed by Slaman–Soskova; rigidity also appears on the expert open-problem list inspected for the original reports. The stronger biinterpretability programme connects it to arithmetic definability.[20, 21]

Paper-and-Lean assessment. One could attack a small automorphism-invariant class instead of all degrees, but proving its invariance is not a solution of G1. The formal burden is likely in coding and definability theorems: representing an order automorphism is easy, whereas proving that enough elements or relations are fixed is not. A weeks-scale attempt should target one genuinely new definability lemma, with its novelty independently established.

11.2 G2. Local rigidity below the halting degree H

Is every automorphism of $(\mathcal{D}_{\leq \mathbf{0}'}, \leq)$ the identity? This is a separate question from G1: an automorphism of the interval need not extend to the full degree structure. Slaman–Soskova relate local rigidity to biinterpretability with first-order arithmetic.[20]

Paper-and-Lean assessment. Restricting the domain to the countable interval below $\mathbf{0}'$ does not turn it into a finite combinatorial problem. A countable coding argument can still require an

extensive oracle-construction library. The principal source used here is older; confirm the exact current local-rigidity status before committing a project.

11.3 Foundations for G3–G5

Let $f : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ be Turing invariant:

$$x \equiv_T y \implies f(x) \equiv_T f(y).$$

A statement holds *on a cone* if it holds for every $x \geq_T z$ for some z . Set

$$f \leq_M g \iff (\exists z)(\forall x \geq_T z) f(x) \leq_T g(x).$$

Martin’s full conjecture is normally formulated in a determinacy setting, such as $\text{ZF} + \text{AD} + \text{DC}$. Borel restrictions provide targets in ordinary classical mathematics. These are not interchangeable foundational formulations.[7]

11.4 G3. Martin’s conjecture, Part I H

In the determinacy setting, must every Turing-invariant f either be constant in degree on a cone or satisfy $x \leq_T f(x)$ on a cone? The Borel-restricted version asks the same dichotomy for Borel f .

Source and status. The general problem is listed in the expert record inspected for the original reports. Lutz–Siskind establish Part I for order-preserving and for measure-preserving functions in the appropriate settings; these subclasses are not still-open versions of the same target. Here measure preservation concerns Martin’s cone measure, not the Lebesgue measure used in M7.[7, 22, 23]

Paper-and-Lean assessment. The difficult step is a uniform structural classification of arbitrary invariant functions, not iteration of a familiar jump construction. For Lean, choose the Borel or internal-set-theoretic statement at the outset. A theorem proved after adding full ambient determinacy to Lean’s ordinary choice principles is not an acceptable formal resolution; see Section 13.9.

11.5 G4. Martin’s conjecture, Part II H

For the Turing-invariant functions above the identity on a cone, is $\leq_M$ a prewellordering, with

$$\text{rank}(x \mapsto f(x)') = \text{rank}(f) + 1?$$

A prewellordering means that the quotient by mutual $\leq_M$ is well-ordered. The foundational and Borel distinctions are as in G3. The prewellordering requirement includes totality as well as well-foundedness after quotienting; well-foundedness alone is weaker.[7]

Paper-and-Lean assessment. A short construction producing two problematic functions would be valuable, but verification would still require exact invariance, cone behaviour, and descriptive complexity. The full positive theorem asks for more than proving that familiar jump iterates form a chain. A formal development of ordinal ranks and the relevant determinacy consequences is a major prerequisite.

11.6 G5. Steel’s uniformity conjecture H

Under determinacy, is every Turing-invariant function f equivalent on a cone to a uniformly Turing-invariant function g defined on a cone? Uniformity means that a single map on pairs

of indices transforms any witnesses of $x \equiv_T y$ into witnesses of $g(x) \equiv_T g(y)$. The index-transforming map is not required to be computable in the definition used by Marks–Slaman–Steel.[7, Conjecture 1.4]

The Borel formulation asks this for Borel f in the ordinary classical setting. In the version used here, one must not add Borelness of the witness g unless it is part of the chosen source formulation. Marks–Slaman–Steel’s Theorem 1.8 relates the Borel uniformity formulation to the Borel Martin programme.[7]

Paper-and-Lean assessment. An accidental extra computability condition on this map would change the problem. The opportunity is an interface theorem between extensional invariance and uniform index control, but there is no identified short Lean dependency chain. A restricted theorem must state exactly which class of invariant functions it covers.

11.7 G6. Universality of Turing equivalence H

Is E_T , Turing equivalence on $2^{\mathbb{N}}$, universal among countable Borel equivalence relations? Explicitly, for every such relation E on a standard Borel space X , must there be a Borel map $F : X \rightarrow 2^{\mathbb{N}}$ with

$$x E y \iff F(x) \equiv_T F(y)?$$

Source and status. Marks–Slaman–Steel discuss this question and its tension with Martin’s conjecture. Their universality theorem for *arithmetic equivalence* does not settle universality for Turing equivalence.[7]

Paper-and-Lean assessment. Borel reductions, countability of equivalence classes, and the exact biconditional all matter. A reduction preserving equivalence in only one direction is insufficient. A potential short counterexample argument is worth exploring conceptually, but a complete formal proof would require substantial descriptive-set-theoretic infrastructure.

11.8 G7. Sacks’s order-embedding conjecture H

Does ZFC prove that every locally countable partial order of cardinality at most the continuum embeds into $(\mathcal{D}, \leq)$? Here locally countable means that every principal down-set is countable, and an embedding e satisfies

$$p \leq q \iff e(p) \leq e(q).$$

Source and status. The expert record lists the conjecture as open. Higuchi–Lutz distinguish the solved height-two case from the difficulties at height three; negative Borel or weak-choice results are not a refutation of the full non-Borel ZFC statement.[24, 25]

The embedding theorem for partial orders of size at most $\aleph_1$ yields the positive case under CH. This does not settle the continuum-sized assertion in ZFC without CH.[26, 24]

Paper-and-Lean assessment. Height three is a meaningful restriction but still carries uncountable global coherence requirements. A finite-poset embedding is not evidence of progress on that obstacle. A formal resolution may require both degree coding and a substantial cardinal/set-theoretic argument; it is not a recommended weeks-scale joint target.

12 Three narrower programmes with better-controlled scope

This section deliberately changes category. The following are proposed research programmes, not three additional literature-certified open problems. A result counts as research only after checking that its exact statement is not already known or an immediate corollary. The aim is to expose a tractable proof boundary, not to rename routine formalization as mathematical novelty.

12.1 R1. A certified bounded lattice-obstruction theorem R

Choose an explicit finite presentation class relevant to L3: for example, join-generated lattices satisfying a fixed, decidable shape condition and avoiding the four named obstructions in Ko’s analysis. Fix the size bound *after* estimating the enumeration cost. The target is a classification theorem of the form

$$\forall L \in \mathcal{C}_{\leq n} \quad \text{Obstruction}(L) \vee \bigvee_{i < r} L \cong L_i,$$

where the L_i are explicitly supplied and the conclusion is a new restriction on viable candidates.

There are three layers to keep separate. First, enumerate finite algebra tables or order relations. Second, certify the lattice laws, the absence or presence of specified sublattices, and isomorphism classes. Third, prove that the resulting obstruction or residual family is relevant to the *degree-theoretic* spectrum target. Only the third layer converts a catalogue into a result about the research problem.

Concrete tool architecture. Use Python or a SAT solver to discover presentations. Export certificates consisting of operation tables, embedding maps, or isomorphism maps. In Lean, express the operation tables on `Fin n`, verify the lattice laws, and check each certificate by kernel-reconstructible finite proofs. For an exhaustive theorem, the verifier must also certify that every admissible input is covered; checking a list of successful examples is not completeness.

A practical route is to enumerate labeled posets by a canonical construction and prove the construction exhaustive, then quotient the result by checked isomorphisms. A faster untrusted generator is acceptable only if its coverage claim is reconstructed by a verified search or an independently checkable exhaustive certificate. Solver failure is not a nonexistence proof.

Novelty gate. Ko already treats important small structural classes. Re-enumerating those classes or finding an example that violates a known obstruction theorem is not a new result. A defensible paper would establish a previously unproved minimality threshold, a new structural reduction, or an infinite-family theorem suggested by the finite search. The best weeks-scale outcome may be the last option: a short induction extracted from a small complete catalogue.

Paper-and-Lean assessment. Among the proposed programmes, this has the most controllable Lean component because finite objects admit explicit certificates. Its mathematical novelty is less certain than M1’s. It should be chosen only after stating the theorem that the computation is intended to prove.

12.2 R2. A new restricted meet-removal theorem R

Select a syntactically explicit family of enriched presentations from L4. The target must include a precise list of preserved joins, common-lower-bound gates, and nonmeet requirements. Attempt a sufficient condition $C(L)$ under which

$$C(L) \implies B(L) = B(T(L)),$$

where T is the correctly encoded meet-removal operation.

One possible proof architecture is a finite combinatorial reduction showing that all uses of a meet can be replaced by a specified gate system, followed by one reusable oracle construction proving that the reduction preserves realizability. This is a proposal for organizing a proof, not a claim that this reduction already works. A counterexample to a proposed sufficient condition is also useful for narrowing the target.

Paper-and-Lean assessment. This is more directly connected to a mathematical open conjecture than R1, but it is less controllable formally. Proceed only if the degree-theoretic step can be reduced to a small, explicit construction. A proof that assumes the full realization theorem it is meant to extend would be circular.

12.3 R3. A new finite bookkeeping theorem for measure constructions R

The recent AEWG work gives a concrete setting in which finite strings, clopen sets, and priority bookkeeping interact.[6] Search for a new abstraction that isolates a quantitative resource invariant: for example, a theorem about cancellation and reassignment of a finite collection of clopen requests under an explicitly bounded number of injuries.

The formal target should quantify over finite prefix-free lists, their dyadic weights

$$w(S) = \sum_{\sigma \in S} 2^{-|\sigma|},$$

and a fully specified transition relation. It should prove a genuinely stronger preservation or budget bound than the lemmas already used in the source. A basic Kraft inequality or the verification of an existing finite-stage lemma is not a new open-problem solution.

A promising implementation discipline is to make the transition system executable while proving an invariant independent of the search strategy. An AI model can propose candidate potential functions and adversarial transition sequences; exhaustive checks on small states can refute an incorrect invariant early. The final theorem must quantify over arbitrary finite executions, and any passage to an infinite oracle construction must be separately proved.

Paper-and-Lean assessment. The finite theorem can be a weeks-scale Lean target. Whether it deserves a conventional research paper depends on a new consequence or a meaningful generalization, not merely cleaner bookkeeping. Completing it does not settle M7 unless the full measure-one and genericity arguments are also supplied.

12.4 Known mathematics that would still be valuable to formalize

A relative oracle-machine enumeration, the general jump, a limit-lemma development, and a carefully engineered finite-injury theorem would improve the prospects of many entries. The Friedberg–Muchnik construction is another substantial infrastructure milestone. None is an open mathematical problem in the sense requested here.

This distinction allows a productive fallback without misreporting success. A project may end with a reusable Lean library and a formalization paper rather than a solution of a new mathematical conjecture. That is a different deliverable, and its value should be stated on its own terms.

13 Lean foundations and verification obligations

13.1 Inspected repositories and reproducible pins

Report A’s audit inspected the `VladimirReshetnikov/ProveIt` repository and the Mathlib file it imports, rather than inferring availability from module names. The inherited source snapshot and environment pins are:

Component	Inspected identifier
ProveIt snapshot	d3b47e6d6cd21efe3a2dad282adc6459c4db1626
Lean toolchain	leanprover/lean4:v4.32.0
Mathlib revision	81a5d257c8e410db227a6665ed08f64fea08e997
Classical degree layer	Computability/TuringDegrees/Lean/

Report A records reading the root toolchain and manifest at the pinned snapshot. It also inspected the coverage ledger on the repository’s then-current main branch. This was a *source audit*: Report A did not build the repository, and the present synthesis has not repeated a Lean build.[11, 27, 9]

13.2 Available foundations versus advertised interfaces

Foundation	What the inspected development provides
Classical set oracles	Total 0/1 characteristic oracles and set reducibility.
Degree quotient and order	Equivalence, quotient, partial order, bottom, and computable-degree characterization.
Join	Even/odd join construction and a join-semilattice interface.
Halting degree	The ordinary halting degree and $\mathbf{0} < \mathbf{0}'$.
Cardinality	Countability of lower cones and degree classes, and continuum-sized degree/upper-cone results, according to the ledger.
General jump	Not supplied by the inspected Lean layer; parameterized jump notions are not a proof of the general jump theorems.
Priority existence arguments	The ledger does not claim Lean proofs of minimal-degree existence, c.e. density, exact-pair existence, or Friedberg–Muchnik.
Pure order consequences	Some implications from minimal degrees, exact pairs, or lack of greatest lower bounds are present; these do not construct the required witnesses.
Rocq material	Some stronger statements are conditional on explicit enumeration or logical principles; they are neither automatically unconditional nor automatically Lean proofs.

The negative statements are scoped to the inspected sources and ledger. They are not a claim that no other public Lean project contains relevant work. Nevertheless, they are enough to reject a planning assumption that the entire classical priority-method toolkit can be imported immediately.[27]

13.3 Reconciling the two software audits

Reports B and C inspect Mathlib documentation and source links associated with commit a218e50f981942cba4fd060faff7cae680805062. This is an additional source-inspection identifier, not a tested dependency pin for Report A’s environment. All three inputs report no Lean compilation. The absence of a theorem from inspected files is evidence about those files, not

proof that no public library contains it. The positive ProveIt evidence in Report A therefore supplements, rather than contradicts, the more cautious Mathlib-only assessments.[1, 2, 3, 10, 28]

13.4 The partial-oracle representation trap

It is unsafe to equate arbitrary partial-map representatives with the classical degrees of their graphs without a theorem. In the inspected interface, ordinary partial computable functions reduce to every oracle. Consider, for example, a partial computable function that returns 0 exactly on the halting set and otherwise diverges. As a partial computation it needs no oracle; its graph, considered as a set with a total membership oracle, is not computable. Graph access answers negative questions about divergence, whereas running the partial function does not. This elementary example explains why the distinction has mathematical content.

For this portfolio, the safest interface is therefore to restrict to total characteristic functions for set oracles, and total natural-valued functions for Baire oracles. Prove that the resulting relative-computability relation agrees with ordinary oracle-machine Turing reducibility. Then transport relations through the set-degree quotient. Merely renaming the existing partial-map quotient or postulating the desired adequacy theorem would not complete a formalization of the classical claims.

13.5 A minimal dependency graph for the padding project

The tower validation project does not need a general priority framework. A reasonable module boundary is the following; these are planned components, not claims about existing Lean declarations.

Layer 1: finite-error combinatorics. Define almost inclusion by finiteness of the set difference, eventual comparison similarly, and infinitude in the ordinary set-theoretic sense. Prove transitivity, closure of finite errors under finite unions and images, and infinitude after removing a finite set. Prove that a set in a finite-fiber region has infinite projection when it is infinite. All of these are independent of computability and should use existing finite-set and order machinery where appropriate.

Layer 2: bounded effective operations. Prove computability of pairing and its inverses, finite conjunctions of column queries, bounded existential tests, and bounded maxima. Establish the specialized facts needed for (3) and (6). The proof should show relative computability from the whole input array without introducing a computable enumeration of the column programs.

Layer 3: tower structures and maps. Represent a tower as an array together with propositions asserting columnwise computability, infinitude or unboundedness, order conditions, and maximality. Define the two padding maps by (2) and (5). Prove the combinatorial properties separately from the computability of the array transformer. This separation makes the critical quantifier issue about f_0 explicit.

Layer 4: reducibility and classical interpretation. Package each transformation as one oracle-computable operator, derive its Medvedev correctness, and then derive the Muchnik and degree-spectrum corollaries. Complete the adequacy bridge to standard total set/function oracles. The final theorem must include this bridge or use a previously verified one; an abstract order-theoretic axiom named “computable” is not an acceptable substitute.

13.6 Representative theorem obligations

The following formulas specify the intended Lean goals without pretending that uncompiled pseudo-code is a finished library:

$$\begin{aligned} \text{WeakTower}(G) &\implies \text{StrictTower}(\text{weakPad}(G)), \\ \text{FunctionTower}(f) &\implies \text{StrictTower}(\text{functionPad}(f)), \\ \text{ComputableIn}(G, \text{weakPad}(G)), \\ \text{ComputableIn}(f, \text{functionPad}(f)). \end{aligned}$$

The last two lines are schematic and must ultimately be strengthened to a single functional independent of the valid input; alternatively, prove that by an explicit shared oracle-program definition. Pointwise existence of a program for each input would prove only the weaker relation.

For the contradiction arguments, separate these two obligations:

$$\text{Computable}(P) \implies \text{Computable}(R), \quad \text{Computable}(P) \wedge \text{Computable}(f_0) \implies \text{Computable}(h).$$

The index for f_0 may occur in the existential computability proof of h ; it must not become an extra oracle supplied to the forward reduction.

13.7 A dependency-oriented architecture

A reusable development should separate five layers. The first is finite coding: programs, strings, stage computations, and finite use. The second is relative computability: oracle semantics, closure operations, effective enumeration, and the parameter theorem. The third is construction infrastructure: approximations, restraint, injury, stabilization, and requirement satisfaction. The fourth is the chosen mathematics: randomness and density for M1–M7, ordinal approximation for L3, or dimension for D1–D4. The fifth is degree-invariant packaging and quotient-level statements.

This is a proposed architecture, not a description of already implemented modules. It prevents a frequent failure mode: proving a theorem about an abstract interface while leaving the crucial fact that ordinary Turing computation implements that interface as an unchecked assumption.

For a priority construction, keep the following obligations visibly distinct:

$$\begin{aligned} &\text{effective finite-stage transition} \\ &\quad \Downarrow \\ &\text{stabilization / true-path argument} \\ &\quad \Downarrow \\ &\text{limit object has the intended complexity} \\ &\quad \Downarrow \\ &\text{every requirement is satisfied} \\ &\quad \Downarrow \\ &\text{degree-level theorem.} \end{aligned}$$

The first implication is not automatic. A program that successfully runs the first million stages has not established stabilization. Nor does totality of a stage transition establish totality of a limit functional.

13.8 Reducing the metric dependency burden

M1 does not necessarily require formalizing every theorem in the HJS paper. One can define the relevant Hausdorff quantity directly and prove the ten exact equalities for a concrete tuple.

For each equality, prove the needed upper and lower estimates from the definitions. General metric classification, all graph realizations, and the range theorem may then be unnecessary dependencies of that *particular* theorem.

However, avoiding a general theorem is not the same as avoiding its difficult content. A lower estimate for a supremum–infimum expression can require a uniform diagonal argument against all oracles of a given degree. The early proof audit should identify those quantifiers explicitly. Hiding them behind a lemma named “distance criterion” does not make the proof shorter.

M3 is less forgiving: to claim the exact diameter, the formal proof must contain both the upper and lower bounds. A conventional paper may cite an established complementary bound, but the original briefs’ end-to-end Lean requirement still needs that bound from a trusted library or a newly formalized proof.

13.9 Determinacy and Lean’s foundations

Full determinacy for all subsets of Baire or Cantor space is not compatible with unrestricted choice. Therefore, a development using Lean’s ordinary classical choice cannot simply add ambient AD as another axiom and call the result a proof of Martin’s conjecture.

There are two coherent routes. Formalize a Borel-restricted statement in ordinary classical mathematics, proving the precise determinacy results actually required. Or formalize a model of the intended set theory and state the determinacy hypothesis *inside that model*. The latter proves an internal theorem under explicitly represented axioms; it does not identify the internal reals with all ambient Lean functions. The article’s G3–G5 assessments include this foundational overhead.[7, 22]

13.10 Acceptance criteria for a genuine joint result

The final statement should use the same objects, ambient order, and quantifier order as the paper. The Lake project should pin its dependencies and build from a clean checkout. All newly used mathematical lemmas must be proved or imported from audited libraries; there must be no `sorry`, `admit`, or custom axiom carrying the desired construction. Inspect `#print axioms` for the principal declarations and document ordinary foundational dependencies such as choice and quotient soundness.

External computation can suggest constructions, enumerate counterexamples to tentative lemmas, and produce certificates. It must not be trusted for the mathematical conclusion unless the stated proof-assistant trust policy explicitly permits that mechanism. For this project’s strongest interpretation, prefer kernel-checkable proof reconstruction or certificate checking to an opaque solver assertion.

An abstract theorem with a typeclass field asserting, for example, that the necessary random degree exists is only conditional until the concrete instance is proved. Similarly, formalizing the implication “HJS theorem plus new lemma implies M1” is useful intermediate work but does not satisfy the full requested deliverable by itself.

13.11 A shared density interface for C1–C3

A useful arithmetic formulation of density zero is

$$\forall k \geq 1 \ \exists N \ \forall n \geq N \quad k |S \cap [0, n]| < n.$$

One may choose $N \geq 1$ to avoid the zero denominator. Prove finite-union closure and the powers-of-two sparsity estimate at this layer, then connect it to the analytic definition. Effective-dense

descriptions need a separate total output type with an explicit omission symbol; nontermination must not be substituted for omission. This interface supports the diagnostic lemmas without pretending that their target constructions have already been formalized.[3]

14 Recent status corrections and tempting nonproblems

Attractive c.e. degrees. Zhao’s preprint of 16 September 2026 reports a theorem of Hirschfeldt–Royer resolving the c.e. attractive-versus-almost-everywhere-dominating characterization positively. The original Hirschfeldt–Royer manuscript was not independently inspected in Report A’s audit; its result is recorded as *reported settled* by that primary source. The residual dispersive-versus-AEWG question is explicitly still open there.[6] Thus the attractive c.e. characterization from HJS Question 9.2 is *not* offered as a fresh short-project target.

Strong minimal pairs. Cai–Liu–Liu–Peng–Yang prove nonexistence of strong minimal pairs of c.e. degrees. Their 2022 paper supersedes the attractive-looking open-problem lead one might obtain from older discussions and contrary claims. This is distinct from L1’s strong minimal cover.[29]

Finite distributive final segments. The distributive case in $\mathcal{D}_2$ is already a theorem of Lempp and coauthors. The broader join-semidistributive question in L5 is the target, not the theorem that supplies its starting point.[17]

Other scope errors. The order-preserving case of Martin Part I is no longer the general open target.[22] Packing-dimension information is not an answer to D1’s Hausdorff question.[19] And proving a missing theorem in a Lean library does not, by itself, make the mathematics new. These distinctions are more consequential for project selection than a list of famous conjecture names.

Tower recommendations have changed within the synthesis. The original search recommendations in Report C are superseded *as project-selection advice* by the existence of the supplied padding proofs. They are not erased from the record, and their simple-coding counterexamples are preserved. The resulting V rating is a validation priority, not an editorial declaration that the literature questions have been settled.

Do not silently strengthen or weaken the question. The recurring distinctions are: weak versus strong reducibility; literal versus upward degree spectra; nonuniform versus uniform descriptions; least versus minimal degrees; a lattice embedding versus a final-segment isomorphism; classical versus effective or relative dimension; source-defined Borel versus determinacy formulations; and a theorem in Lean’s ordinary ambient mathematics versus a theorem internal to a model of set theory. The definitions in Sections 2–10 fix these choices for the unified text.

15 A unified research and verification protocol

The following sequence integrates the stop/go criteria of all three reports. It is an experimental protocol, not a schedule or a promise that a novel result will appear within a fixed period.

15.1 Fix the statement before building infrastructure

For the tower track, first agree on the exact definitions, array representation, optional $G_0 = \mathbb{N}$ convention, and reduction direction. Ask whether the proposed arguments address the intended source questions and whether equivalent constructions already occur in the literature. For an unresolved target, choose *one* of M1, M3, C1, or a precisely stated restriction, and write its complete quantifiers before selecting a proof assistant interface. The first formal pilot should test the semantic bottleneck, not merely establish notation.

For example, the tower pilot is bounded projection and maximum computability with the right uniformity; M1's pilot is a genuinely usable distance estimate; C1's pilot separates coarse descriptions from Turing noncomputability. An early failure to prove these interfaces is evidence to revise the project, not permission to turn the missing result into an axiom.

15.2 Interleave proof search, counterexample search, and formalization

A proposed lemma should be attacked independently before it becomes a large dependency. Finite examples can falsify combinatorial claims, and model-assisted searches can suggest maps or coding invariants. They do not prove stabilization, maximality, randomness, or the absence of an oracle computation. For a tower separation one would need a single source object computing no target object; showing that one natural map fails is much weaker. For a coarse-class counterexample, a noncomputable representative alone says almost nothing, as Proposition 5.1 shows.

Keep a conventional proof and a theorem-obligation ledger in parallel. Record every imported theorem, the precise formal statement intended to represent it, and whether an implementation is proved, imported, or still missing. If an argument uses an abstract interface, separately establish its concrete instance for ordinary oracle computation. A change of quantifier order must be reflected in both the paper and the code.

15.3 Choose the end product honestly

There are three legitimate but different outcomes: a new mathematical result with a complete formal proof; a novel conventional argument whose formalization remains incomplete; or a formalization of known mathematics that creates reusable infrastructure. A fourth useful outcome is a sharply diagnosed failed approach with proved limitations, but it must not be described as a solution. The present unified report is a synthesis and proof-proposal record; it does not claim any of those completed research outcomes on the basis of document compilation.

For a genuine paper-and-Lean solution, require a clean pinned build, the same semantic statement in paper and code, all substantive dependencies discharged, and a documented axiom audit. A known theorem used without its formal proof is a clear boundary on the formal result, not a hidden convenience. For a bounded classification, a list of verified cases also needs a completeness argument showing that every eligible finite object was considered.

15.4 Combined recommendation

The first choice for an immediate verification project is the tower-padding package, because it has explicit maps, bounded computations, and short maximality arguments already available. The first choices for fresh unresolved work are M1, M3, and C1, conditional on a small dependency path. The best scope-controlled alternatives are R1–R3 after a novelty check. Global structural conjectures remain mathematically valuable targets, but the report identifies no comparably short complete formal route for them. These priorities concern present proof shapes and infrastructure,

not limits on possible discovery.

A Source-to-unified crosswalk

The original identifiers are local to each input. The table accounts for every numbered source entry; R1–R3 are Report A’s separate programmes. Splitting Report C’s P14 into D3 and D4 preserves its classical and relative-effective variants. A crosswalk row does not assert equivalence of every refinement contained in the two entries. The machine-readable counterpart is `source_crosswalk.csv`.

Input	Original ID	Unified topic(s)
Report A	M1	M1
Report A	M2	M2
Report A	M3	M3
Report A	M4	M4
Report A	M5	M5
Report A	M6	M6
Report A	M7	M7
Report A	G1	G1
Report A	G2	G2
Report A	G3	G3
Report A	G4	G4
Report A	G5	G5
Report A	G6	G6
Report A	G7	G7
Report A	L1	L1
Report A	L2	L3
Report A	L3	L4
Report A	L4	L5
Report A	D1	D1; D2 (related variant)
Report A	D2	D4
Report A	P1	R1
Report A	P2	R2
Report A	P3	R3
Report B	P01	G1
Report B	P02	L2
Report B	P03	G7
Report B	P04	G3
Report B	P05	G4
Report B	P06	G5
Report B	P07	G6
Report B	P08	Q1
Report B	P09	Q2
Report B	P10	Q3
Report B	P11	T2
Report B	P12	T1
Report B	P13	T3
Report B	P14	T4
Report B	P15	K1
Report B	P16	K2
Report B	P17	K3
Report B	P18	K4
Report B	P19	N1
Report B	P20	N2
Report C	P01	T1
Report C	P02	T2

Input	Original ID	Unified topic(s)
Report C	P03	T3
Report C	P04	T4
Report C	P05	C1
Report C	P06	C2
Report C	P07	C3
Report C	P08	Q3
Report C	P09	Q2
Report C	P10	Q1
Report C	P11	K1
Report C	P12	D1
Report C	P13	D2
Report C	P14	D3 and D4
Report C	P15	G1
Report C	P16	G3
Report C	P17	G4
Report C	P18	G5
Report C	P19	G6
Report C	P20	G7

B Evidence and software ledger

The primary references below replace three overlapping bibliographies. Their exact versions matter: the McDonald and Zhao items are 2026 preprints; the workshop lists are dated collections of questions, not proofs that no solution exists; older sources for local rigidity and strong minimal covers need renewed specialist confirmation. Report B’s sharpened $\text{Diag} = \text{PA}$ question, the decidability form of finite-lattice classification, and the strongest binary-representative bound remain explicitly scoped components or refinements.

Evidence group	Provenance and limits
Tower definitions and questions	McDonald v1, 20 July 2026; targeted wording and question check for this edition. The proposed proofs come from Report B, not from that preprint.
Coarse and effective-dense questions	Gerdes v1, 9 August 2025, Questions 5 and 7; exact distinction between least representatives and low binary representatives retained.
Metric geometry	HJS preprint and journal record inherited from Report A; Zhao v1, 16 September 2026, targeted status check for M7 and the reported attractive-degree result.
Local classes and categoricity	Tianyuan 2025 list, together with LMNS and Badaev et al.; broad classification requests distinguished from narrower conjectures.
Randomness	Tianyuan 2026 v2, 31 August 2026, Problem 1.4(1)–(2), inherited from Report B.
Lattices and dimension	Ko, Lempp et al., and Song–Yu, at the versions in the bibliography; inherited problem and theorem assessments.
Global structure	Marks–Slaman–Steel, Lutz–Siskind, Slaman–Soskova, Higuchi–Lutz, and the expert registry; foundational variants kept separate.

Evidence group	Provenance and limits
Lean and ProveIt	Dated source inspection only. No original or unified-edition Lean build is reported; the two Mathlib identifiers are not conflated.
Document and finite checks	The unified LaTeX was compiled, references and layout were checked, and both inherited Python finite-check programs were rerun. These checks have a narrower scope than the mathematical and Lean claims.

C Included finite checks and reproducibility

The archive contains a self-contained LaTeX source, its compiled PDF, build instructions, the source crosswalk, and two inherited Python programs with fresh outputs. Font binaries and LaTeX intermediate files are not included.

Padding checks. `checks/check_padding.py`, from Report B, checks finite instances of the weak-tower and function-tower inclusions and error-projection bounds, including exceptional rows and points outside the triangular region. Its default run uses seed 20260917, exhaustively checks finite configurations, and adds deterministic pseudorandom instances. The detailed counts are in `checks/padding_results.txt`.

Metric and graph checks. `checks/finite_checks.py`, from Report A, enumerates all orientations of the five-cycle, all five-vertex distance assignments using $1/2$ and 1 , and a complement isomorphism. Its output is `checks/finite_checks_results.json`. These computations check the finite combinatorial observations in the discussion of M1; they do not realize a graph by Turing degrees.

Trust boundary. Neither script checks infinite pseudointersections, actual oracle reducibility, descriptive complexity, source-author intent, novelty, or a Lean kernel proof. The infinite arguments in Section 4 remain conventional proof proposals to be reviewed. Compilation of this report certifies only that the document builds, not that its mathematical conclusions have been mechanically established.

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